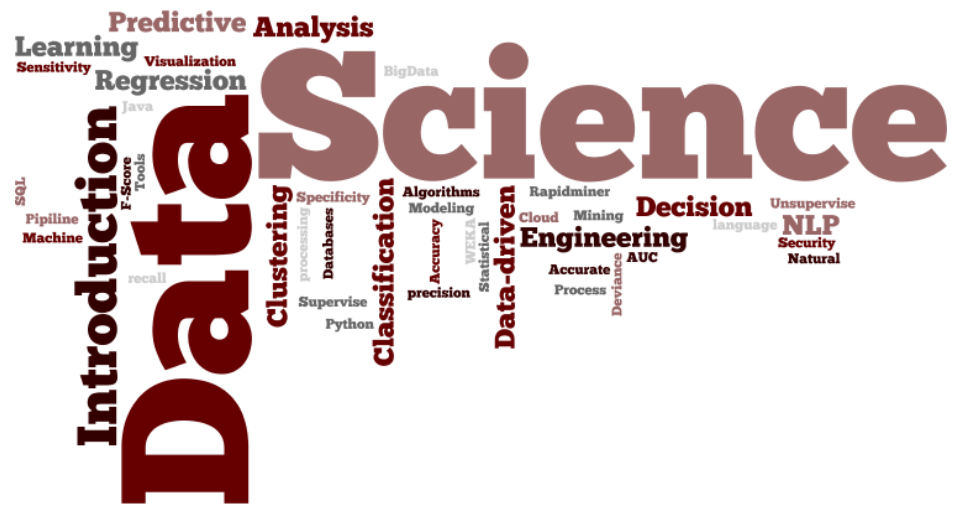




9.3 เครื่องมือวิทยาการข้อมูล: Rapidminer

Rapidminer Auto Model

ดำรงฤทธิ์ เศรษฐศิริโชค

Auto Model

Rapidminer Auto Model



Rapidminer Auto Model

Repository

Import Data

Training Resources (connected)

Samples

data

Deals

Deals-Testset

Golf

Golf-Testset

Iris

Labor-Negotiations

Views: Design Results Turbo Prep **Auto Model** More

Find data, operators...etc

All Stu

Auto Model

Load Data

Select Task

Prepare Target

Select Inputs

Model Types

Results

RESTART

BACK

NEXT

Predict

Want to predict the values of a column?

Clusters

Want to identify groups in your data?

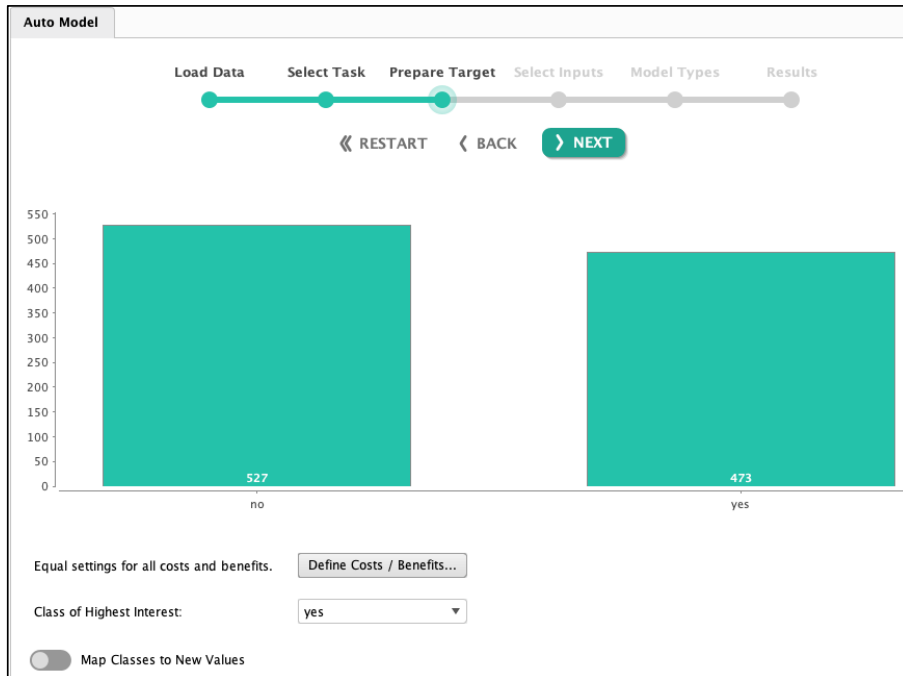
Outliers

Want to detect outliers in your data?

Age Number	Gender Category	Payment Method Category	Future Customer Category
64	male	credit card	yes
35	male	cheque	yes
25	female	credit card	yes
39	female	credit card	no
39	male	credit card	yes
28	female	cheque	no
21	female	credit card	yes
48	male	credit card	yes
70	female	credit card	no
36	male	credit card	yes

1,000 rows - 4 columns (3 nominal, 1 numerical)

Rapidminer Auto Model



Auto Model

Load Data Select Task Prepare Target **Select Inputs** Model Types Results

« RESTART < BACK > NEXT

Selected: 3 / Total: 3

☒ Select All ☒ Deselect All

Selected	Stat... ↑	Quality	Name	Correlation	ID-ness	Stability	Missing	Text-ness
☒	●		⚠ Age	28.17%	?	2.90%	0.00%	0.00%
☒	●		⚠ Gender	22.51%	0.20%	55.00%	0.00%	2.24%
☒	●		Payment Method	16.75%	0.30%	65.20%	0.00%	25.70%

Rapidminer Auto Model

Auto Model

Load Data

Select Task

Prepare Target

Select Inputs

Model Types

Results

« RESTART

< BACK

▶ RUN

Execute on:

Local Computer
(this machine)

Local Computer
(this machine)

AI Hub Repository
(http://20.211.161.39:8080/)

Queue:

Select Folder for Storing Results

The results of this run will be stored in the folder selected below. We recommend to use an empty folder in the selected AI Hub repository.

▶ Local Repository (Legacy)

▶ Temporary Repository (Legacy)

▶ AI Hub Repository (disconnected)

☒ Naive Bayes

☒ Generalized Linear Model

☒ Use Regularization ☐ Calculate p-Values

☒ Logistic Regression

☒ Fast Large Margin

☒ Automatically Optimize

☒ Deep Learning

☒ Decision Tree

☒ Automatically Optimize Maximal Depth: 20

☒ Random Forest

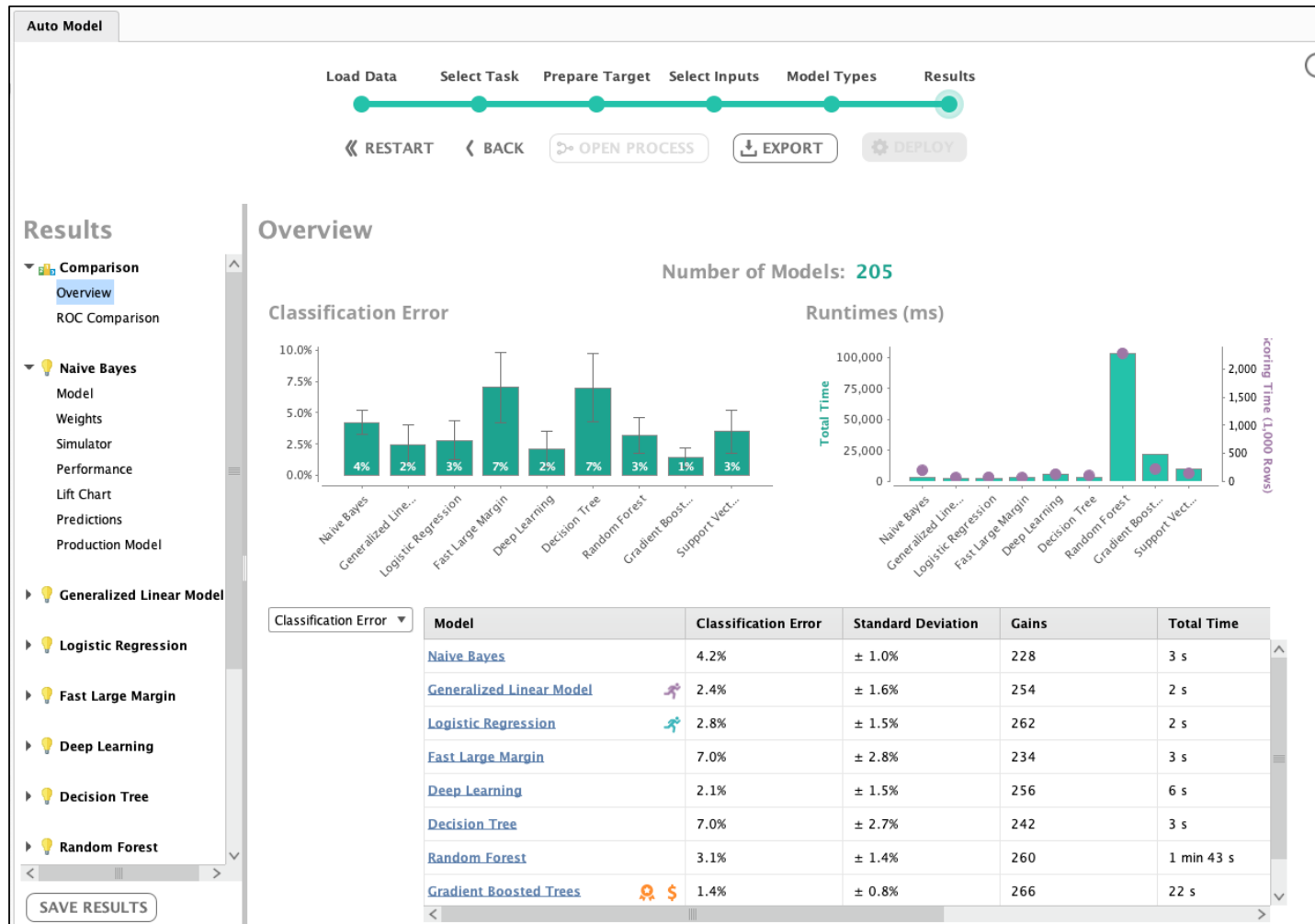
☒ Automatically Optimize Number of Trees: 20 Maximal Depth: 20

☒ Gradient Boosted Trees

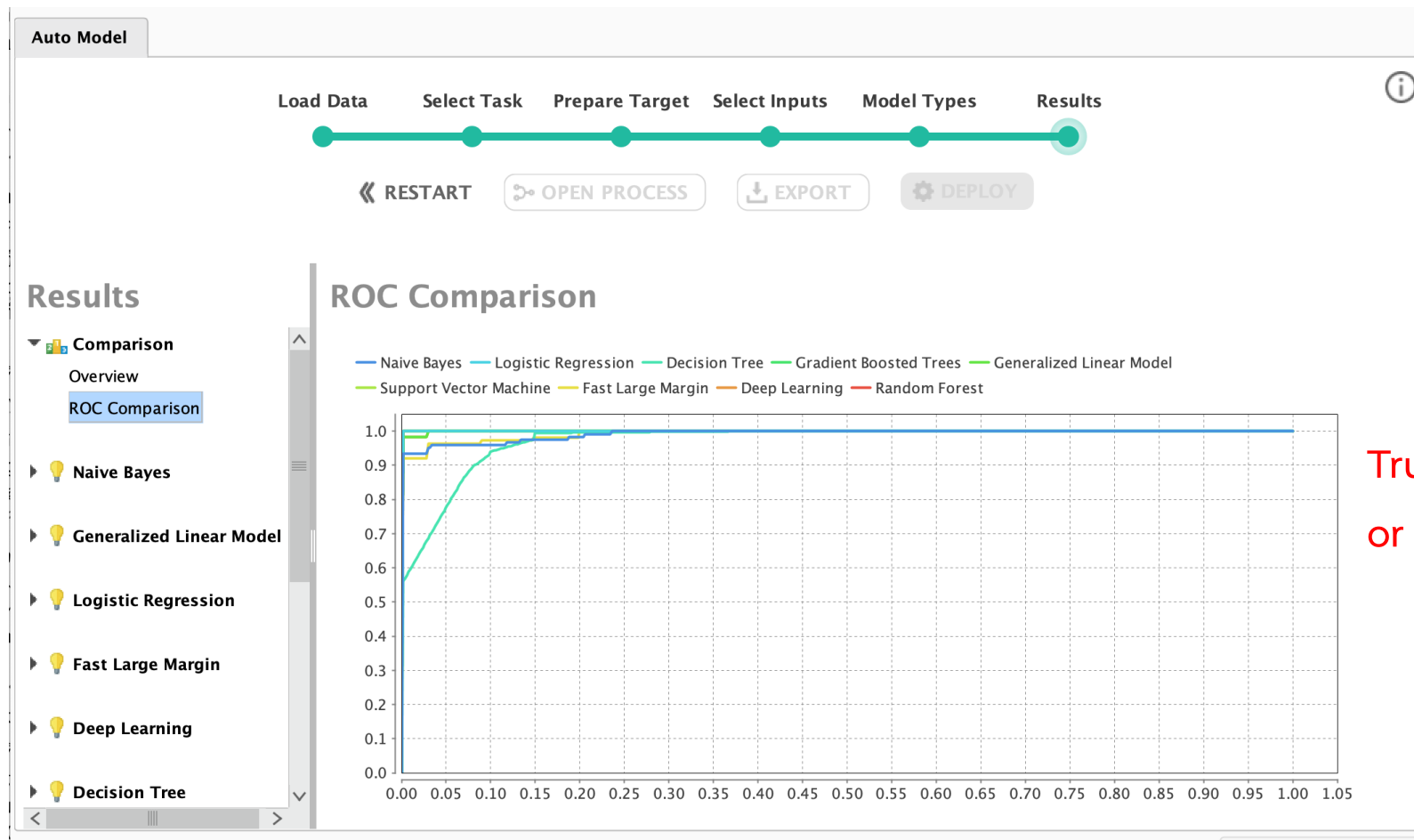
☒ Automatically Optimize Number of Trees: 20 Maximal Depth: 20 Learning Rate: 0.0

☒ Support Vector Machine

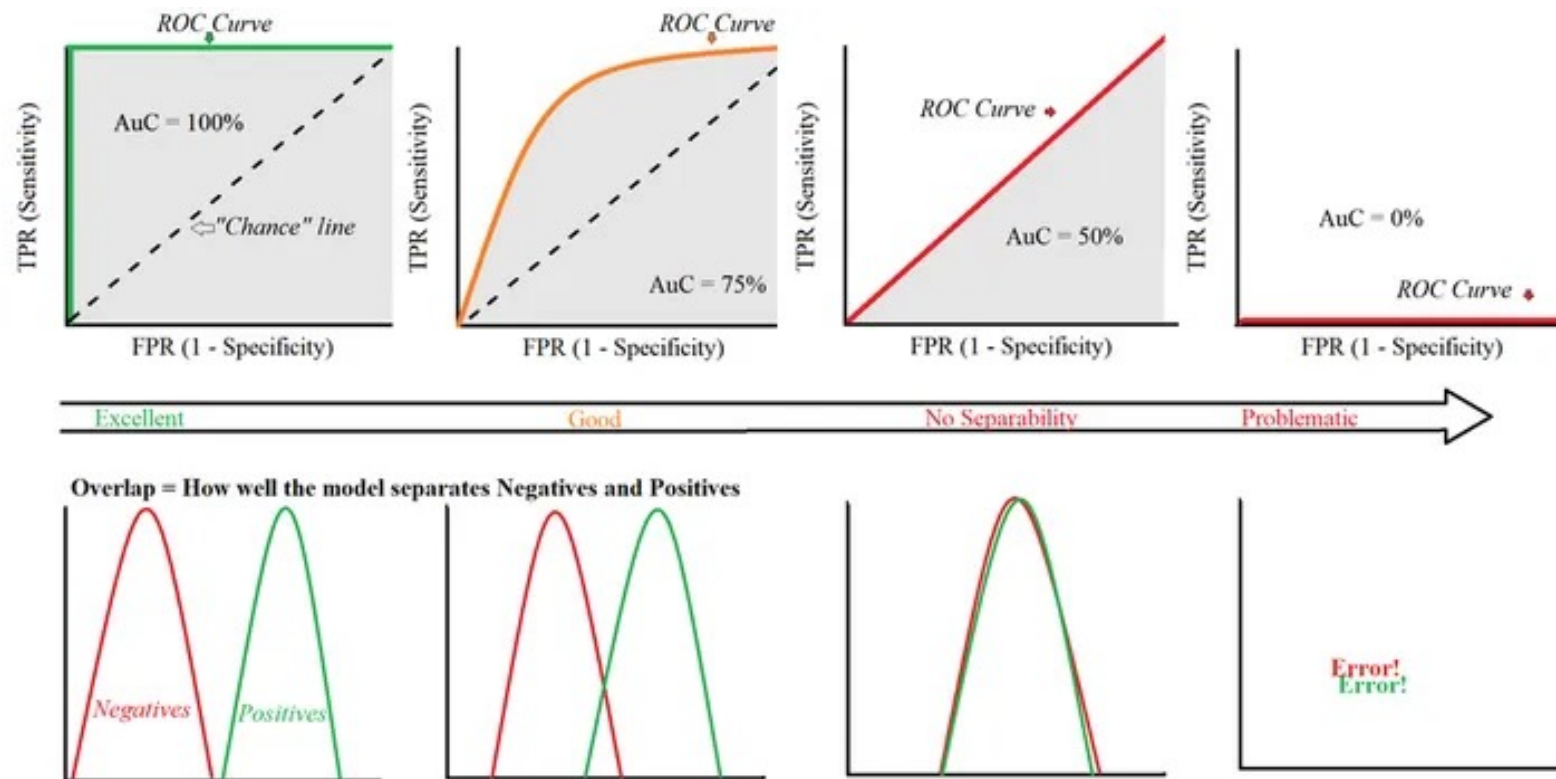
Rapidminer Auto Model – Comparison Overview



Rapidminer Auto Model – ROC



Rapidminer Auto Model – ROC



Rapidminer Auto Model – Model

Results

Production Model

Decision Tree

Model

Weights

Simulator

Performance

Lift Chart

Optimal Parameters

Predictions

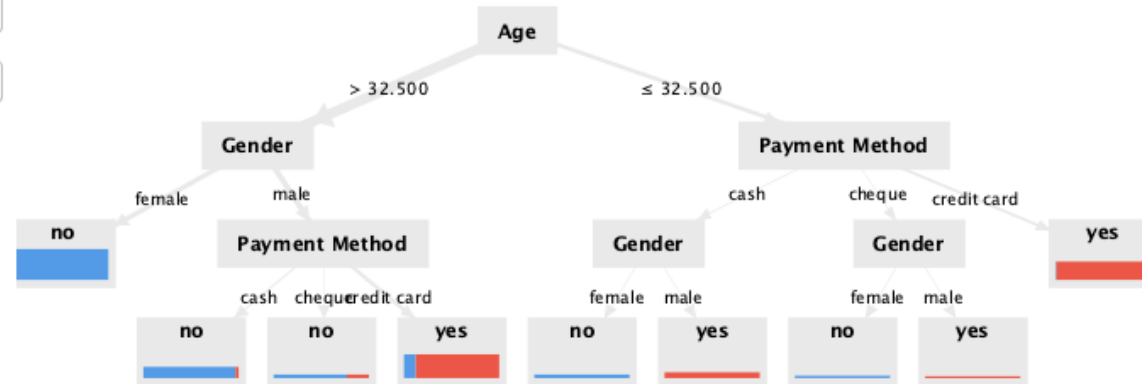
Production Model

Random Forest

Gradient Boosted Trees

Support Vector Machine

Decision Tree – Model



Rapidminer Auto Model – Performance

Auto Model

Load Data

Select Task

Prepare Target

Select Inputs

Model Types

Results

« RESTART

◀ BACK

↻ OPEN PROCESS

📄 EXPORT

⚙️ DEPLOY

Results

Decision Tree

Model

Weights

Simulator

Performance

Lift Chart

Optimal Parameters

Predictions

Production Model

Random Forest

Gradient Boosted Trees

Support Vector Machine

General

Data

Statistics

Weights by Correlation

Correlation

SAVE RESULTS

Decision Tree – Performance

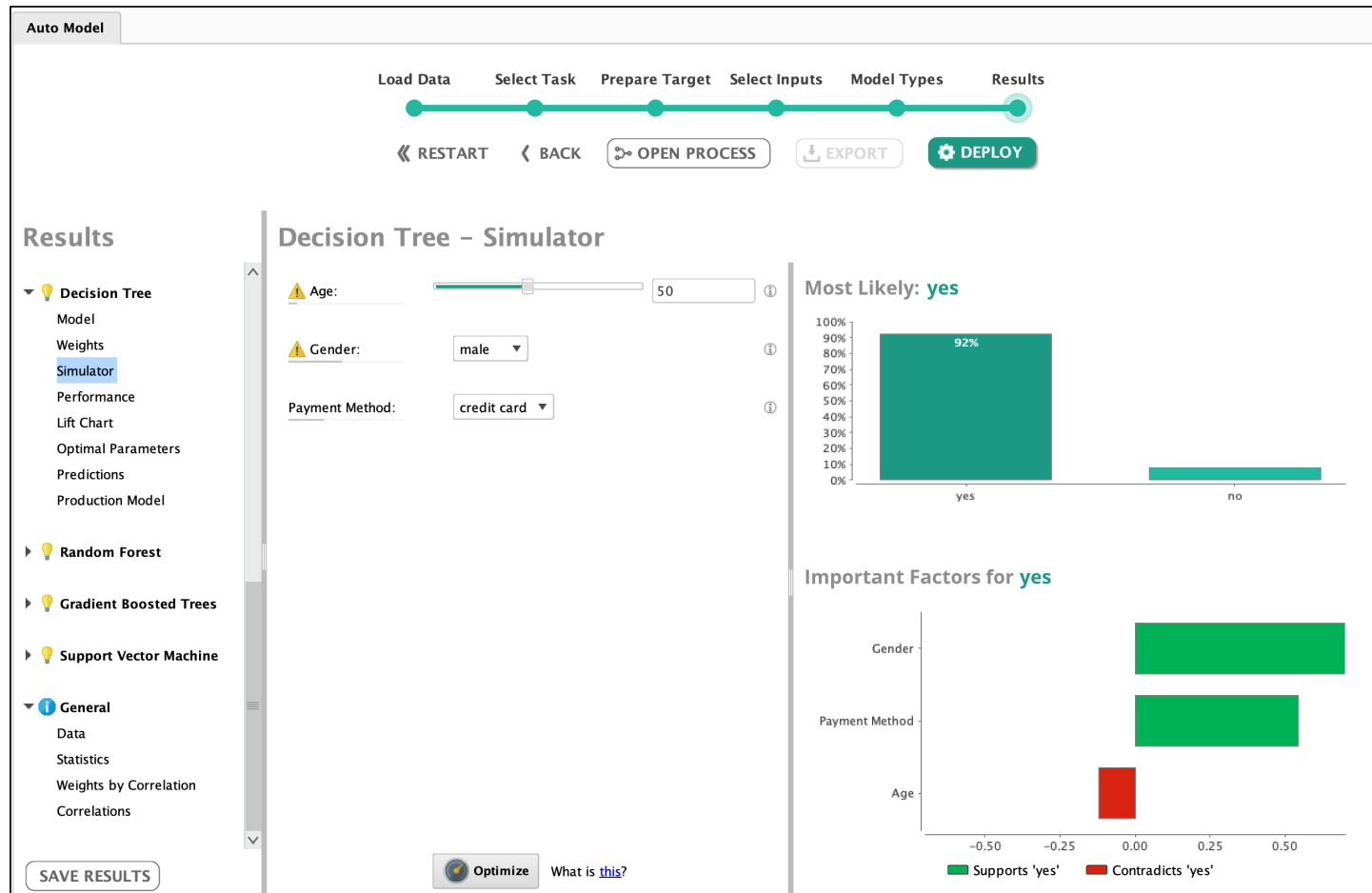
Performances

Criterion	Value	Standard Deviation
Accuracy	93.0%	± 2.7%
Classification Error	7.0%	± 2.7%
AUC	97.4%	± 1.2%
Precision	89.4%	± 4.8%
Recall	97.3%	± 2.7%
F Measure	93.1%	± 2.8%
Sensitivity	97.3%	± 2.7%
Specificity	89.1%	± 3.7%

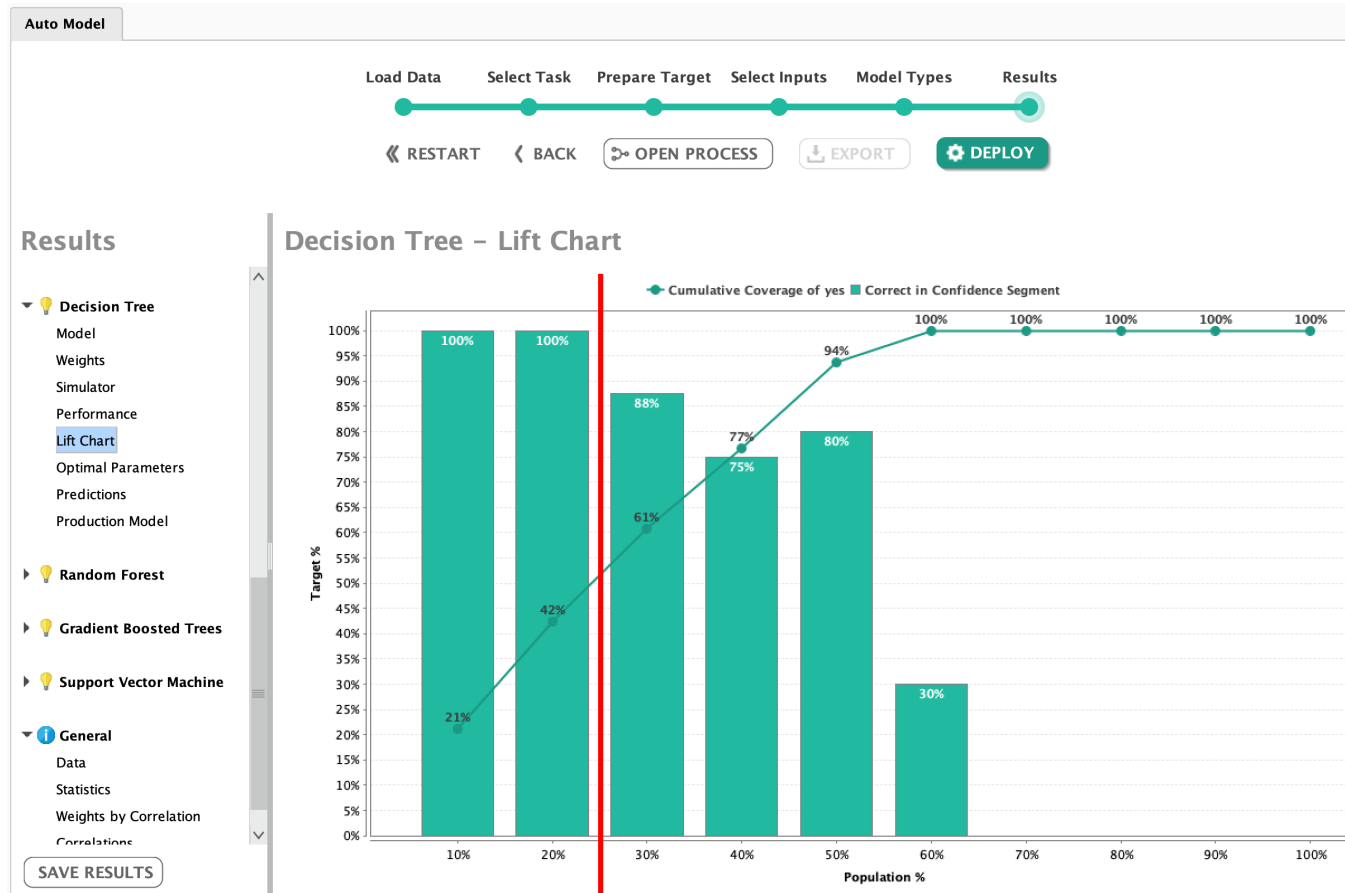
Confusion Matrix

	true no	true yes	class precision
pred. no	129	4	96.99%
pred. yes	16	137	89.54%
class recall	88.97%	97.16%	

Rapidminer Auto Model – Simulator

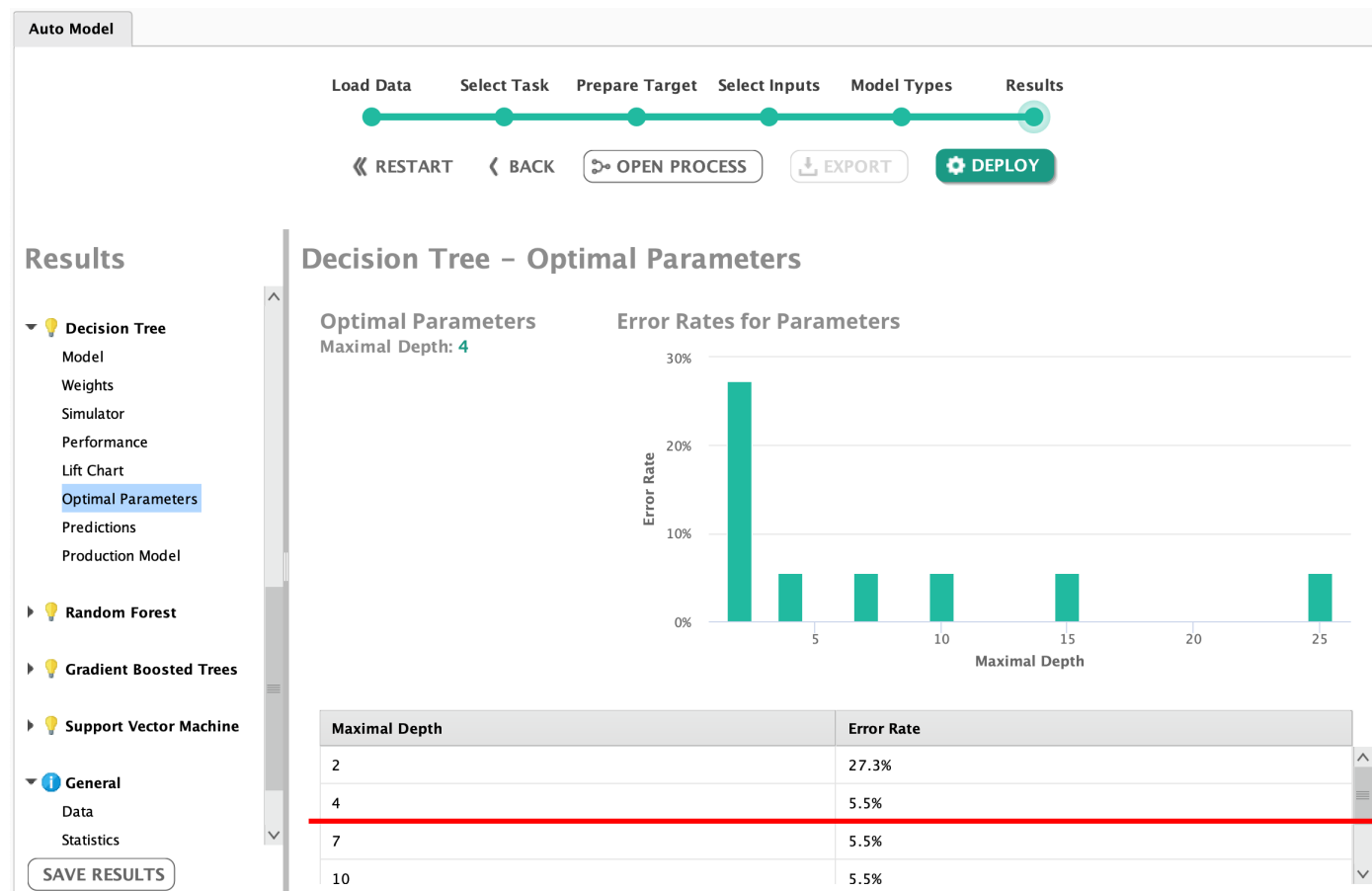


Rapidminer Auto Model – Lift Chart



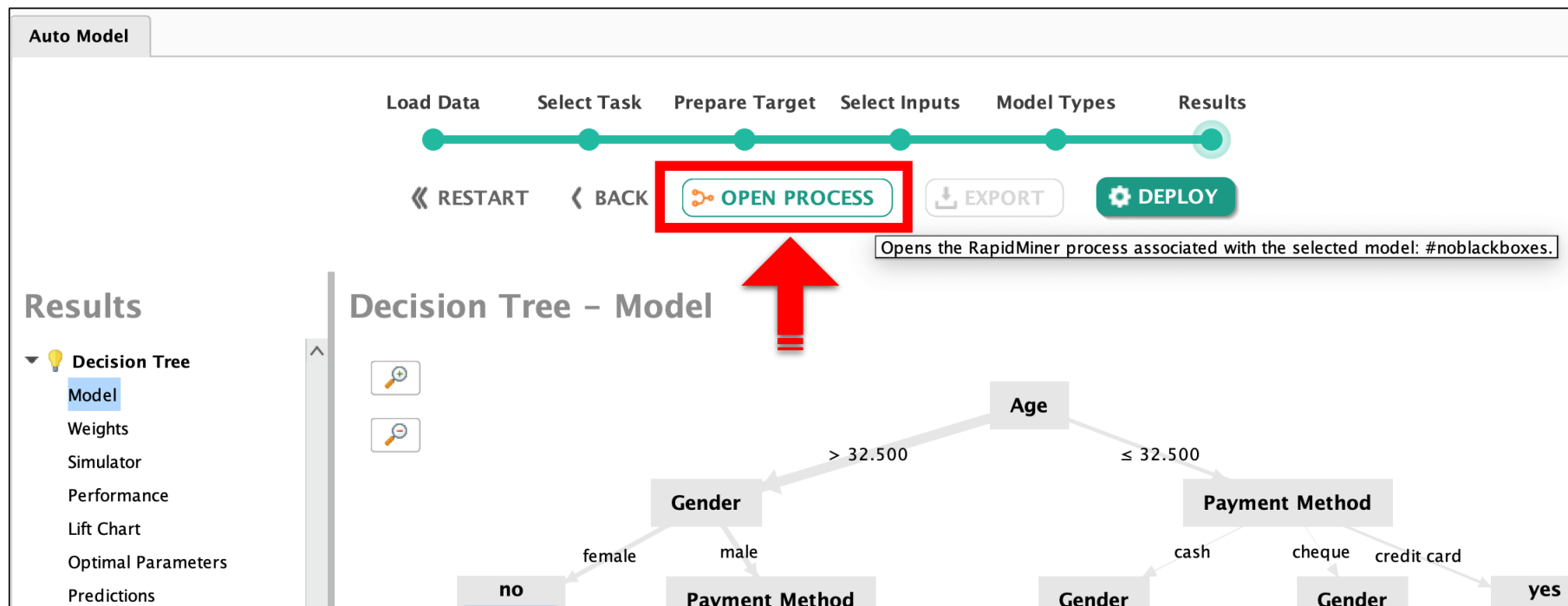
First 20% (42% of class “yes”) → 100% Confidence

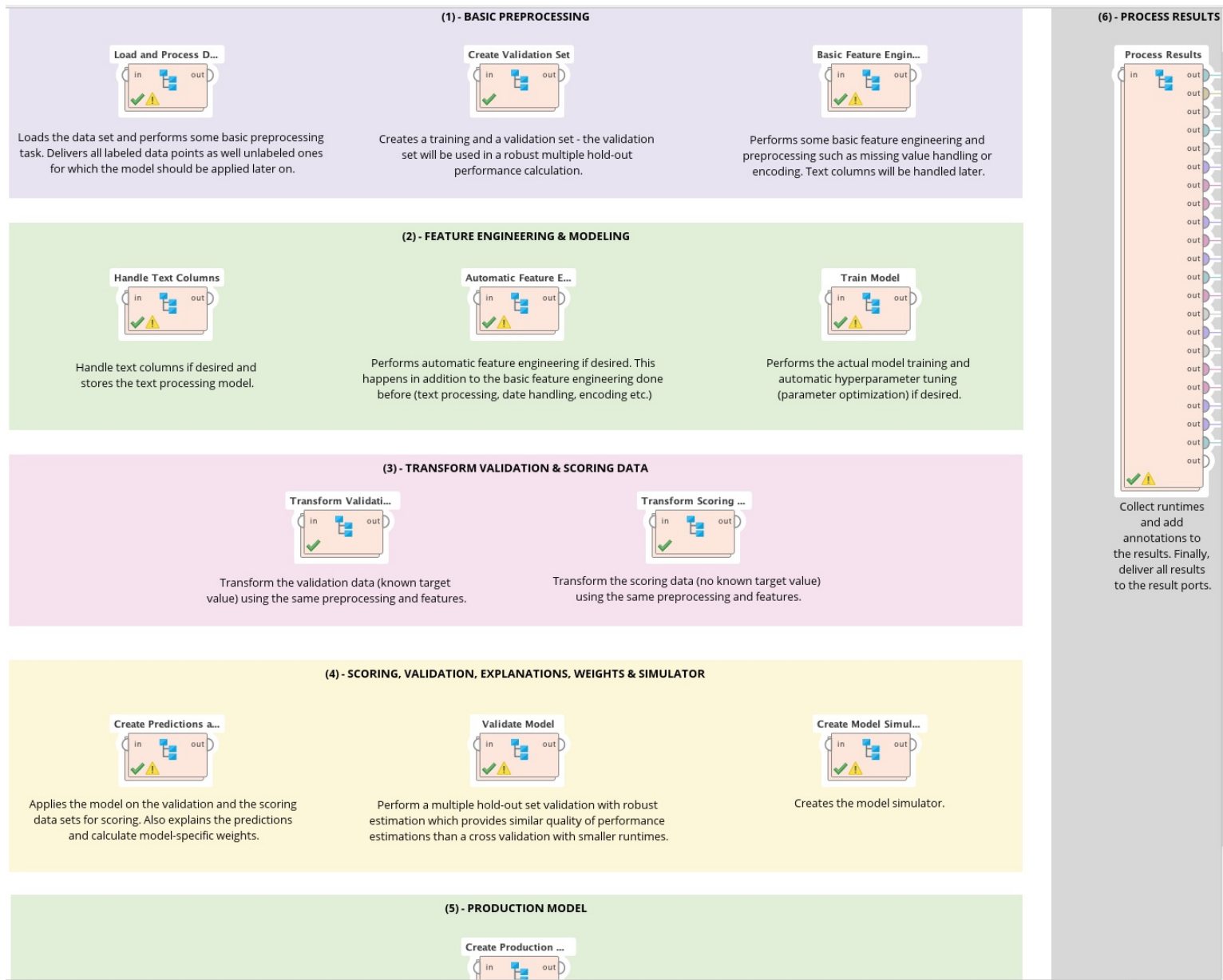
Rapidminer Auto Model – Optimal Parameters



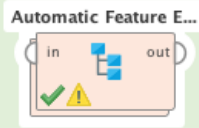
Rapidminer Auto Model – Drill Down

- Process inside

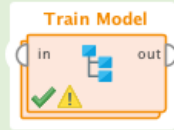




(2) - FEATURE ENGINEERING & MODELING



Performs automatic feature engineering if desired. This happens in addition to the basic feature engineering done



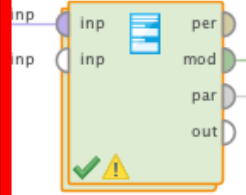
Performs the actual model training and automatic hyperparameter tuning

Process

Process ▶ Train Model ▶ Optimize? ▶

Selection 1

Find Optimal Parameters



Find optimal parameters.

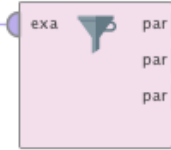
Remember Optimal...



Remember optimal parameters.

Selection 2

Split Data (4)



Create calibration data.

Model



Parameters

Find Optimal Parameters (Optimize Parameters (Grid))

Edit Parameter Settings...

error handling on error

Select Parameters: configure operator

Select Parameters: **configure operator**
Configure this operator by means of a Wizard.

Operators	Parameters	Selected Parameters
Validation PO (Cross Validation)		Model PO.maximal_depth
Split Data (3) (Split Data)		
Model PO (Decision Tree)		
Calibrate PO (Rescale Confidence)		
Apply Model PO (Apply Model)		
Performance PO (Performance (CI))		
Log Performances (Log)		

Grid/Range

Min	Max	Steps	Scale
0.0	0.0	0	linear

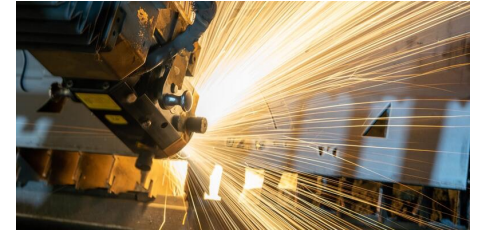
Value List

Value List	Buttons
	+ 2 4 7 10 15 25

Q & A

- Try Auto ML – housing dataset

Auto Model LAB



- Use case: **Machine Predictive Maintenance [Classification]**
- <https://www.kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification>
- Dataset: **MF.csv**

Turbo Prep

Data Sets

+ LOAD DATA

MF
//Local Repository/data/MF
Rows: 10,000
Columns: 7
Last Change: None

ADD NEW DATA SETS ON THE LEFT. DETAILS FOR THE SELECTED DATA ARE SHOWN BELOW. YOU CAN CHANGE THE DATA WITH THE FOLLOWING ACTIONS. ⓘ

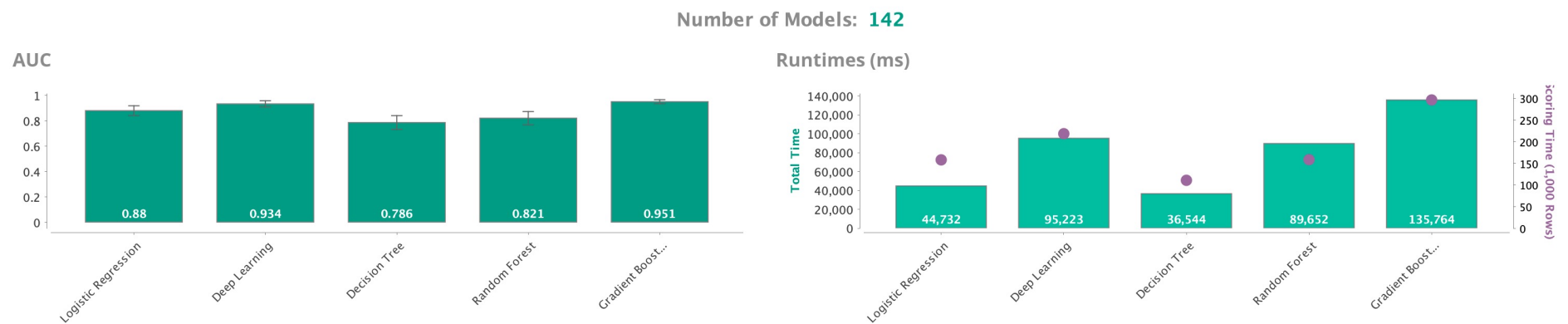
✂ TRANSFORM ✏ CLEANSE 📊 GENERATE ∑ PIVOT ➡ MERGE

MODEL CHARTS CREATE PROCESS HISTORY ...

Type Category	Air temperature [K] Number	Process temperatur... Number	Rotational speed [rp... Number	Torque [Nm] Number	Tool wear [min] Number	MF Category
M	298.100	308.600	1551	42.800	0	Normal
L	298.200	308.700	1408	46.300	3	Normal
L	298.100	308.500	1498	49.400	5	Normal
L	298.200	308.600	1433	39.500	7	Normal
L	298.200	308.700	1408	40	9	Normal
M	298.100	308.600	1425	41.900	11	Normal
L	298.100	308.600	1558	42.400	14	Normal
L	298.100	308.600	1527	40.200	16	Normal
M	298.300	308.700	1667	28.600	18	Normal

Auto Model – Machine Predictive Maintenance

Overview



AUC ▼

Model	AUC	Standard Deviation	Gains	Total Time	Training Time (1,00...	Scoring Time (1,000 ...	Deploy
Logistic Regression	0.88	± 0.038	-42	45 s	20 ms	158 ms	Deploy
Deep Learning	0.934	± 0.024	-8	1 min 35 s	780 ms	219 ms	Deploy
Decision Tree   	0.786	± 0.054	22	37 s	11 ms	111 ms	Deploy
Random Forest	0.821	± 0.052	-2	1 min 29 s	8 ms	159 ms	Deploy
Gradient Boosted Trees 	0.951	± 0.016	10	2 min 15 s	216 ms	297 ms	Deploy

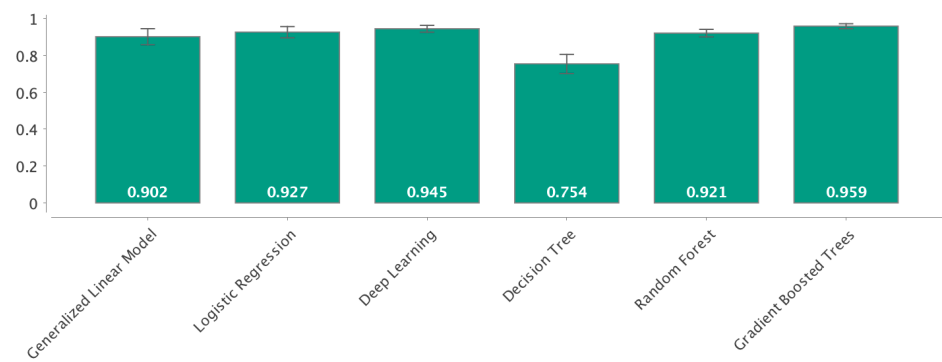
Auto Model – Machine Predictive Maintenance

With feature engineering interaction term

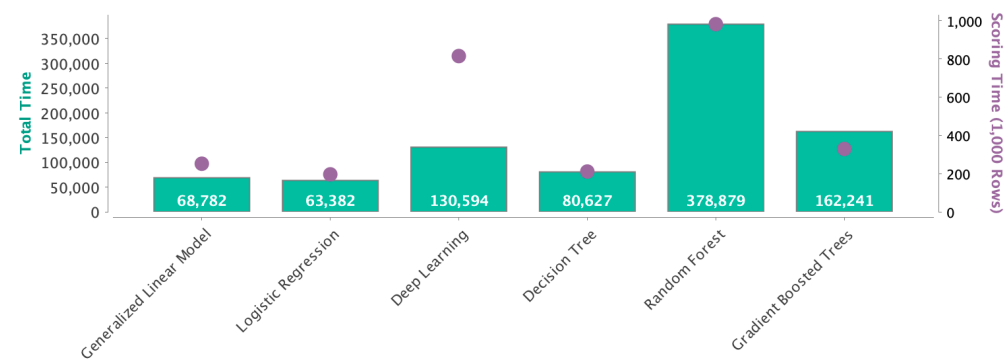
Overview

Number of Models: 144

AUC



Runtimes (ms)



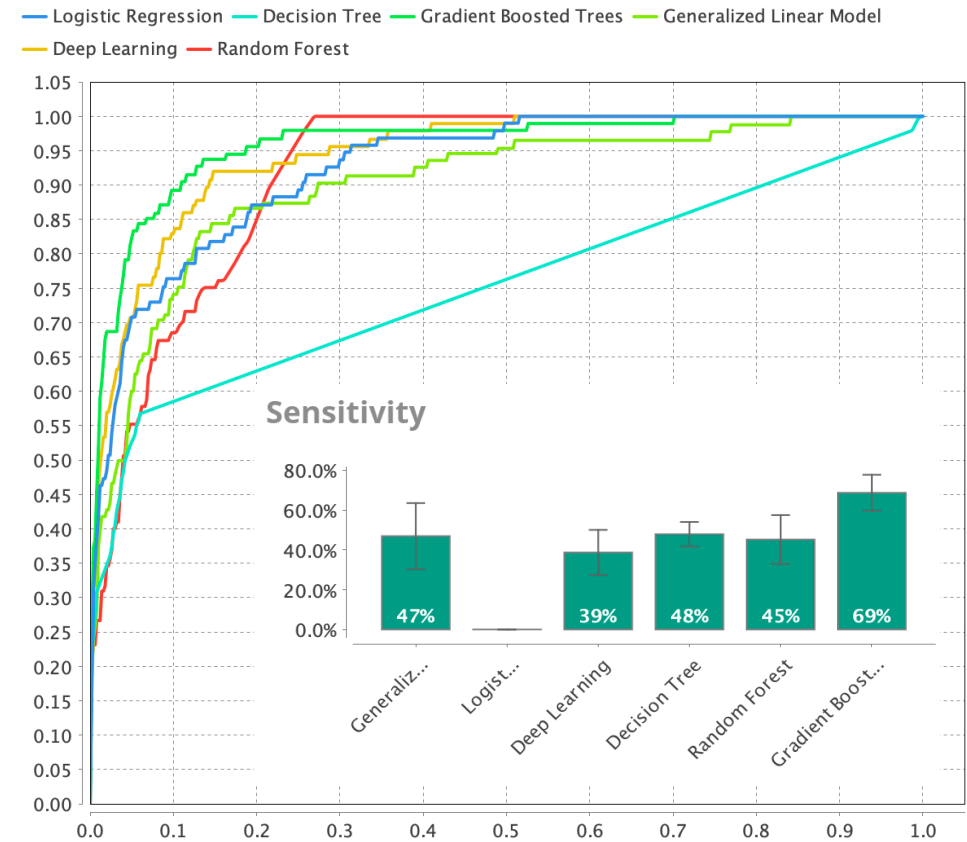
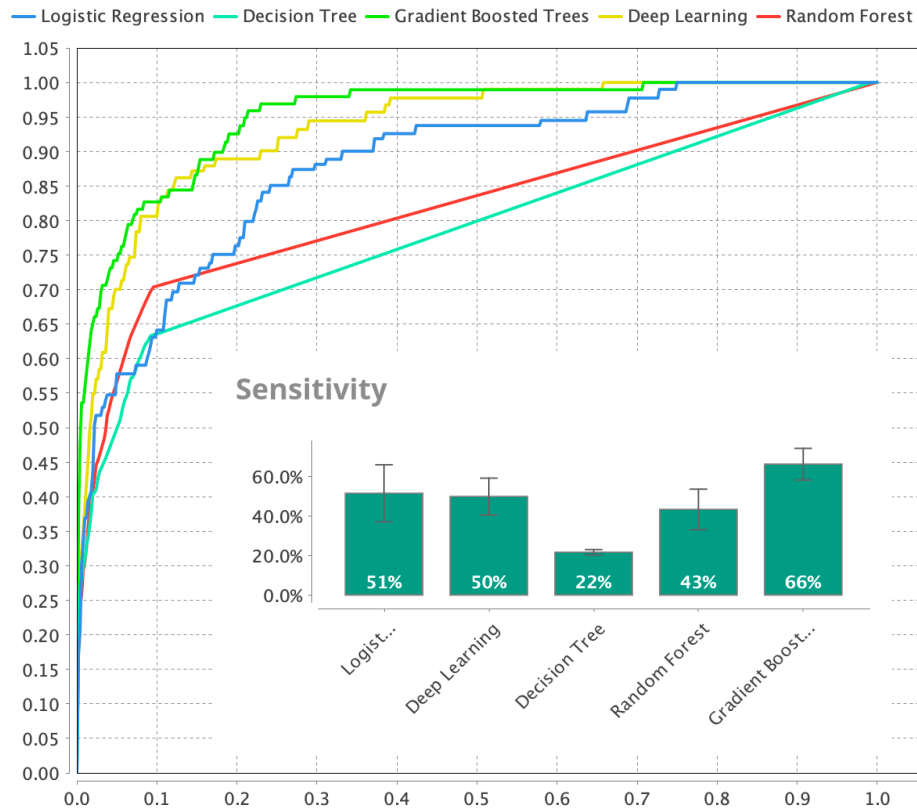
AUC ▼

Model	AUC	Standard Deviation	Gains	Total Time	Training Time (1,000...	Scoring Time (1,000 ...	Deploy
Generalized Linear Model	0.902	± 0.044	-38	1 min 8 s	20 ms	252 ms	Deploy
Logistic Regression	0.927	± 0.031	0	1 min 3 s	22 ms	196 ms	Deploy
Deep Learning	0.945	± 0.019	28	2 min 10 s	822 ms	815 ms	Deploy
Decision Tree	0.754	± 0.052	-90	1 min 20 s	21 ms	211 ms	Deploy
Random Forest	0.921	± 0.021	-122	6 min 18 s	89 ms	982 ms	Deploy
Gradient Boosted Trees	0.959	± 0.013	-64	2 min 42 s	163 ms	329 ms	Deploy

Auto Model – Machine Predictive Maintenance

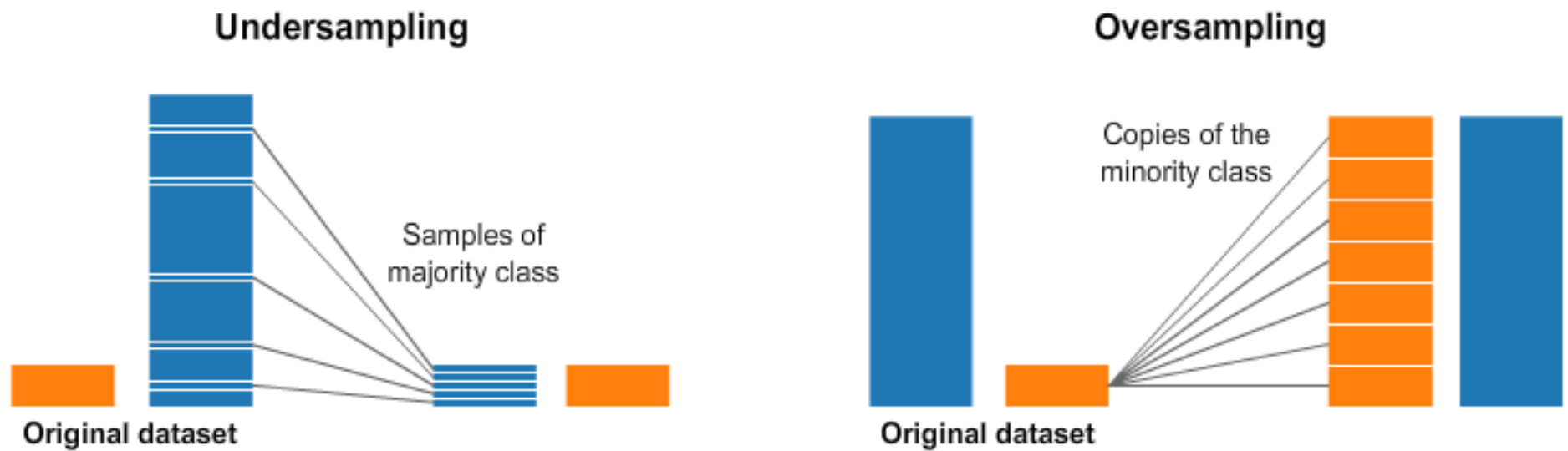
ROC Comparison

ROC Comparison



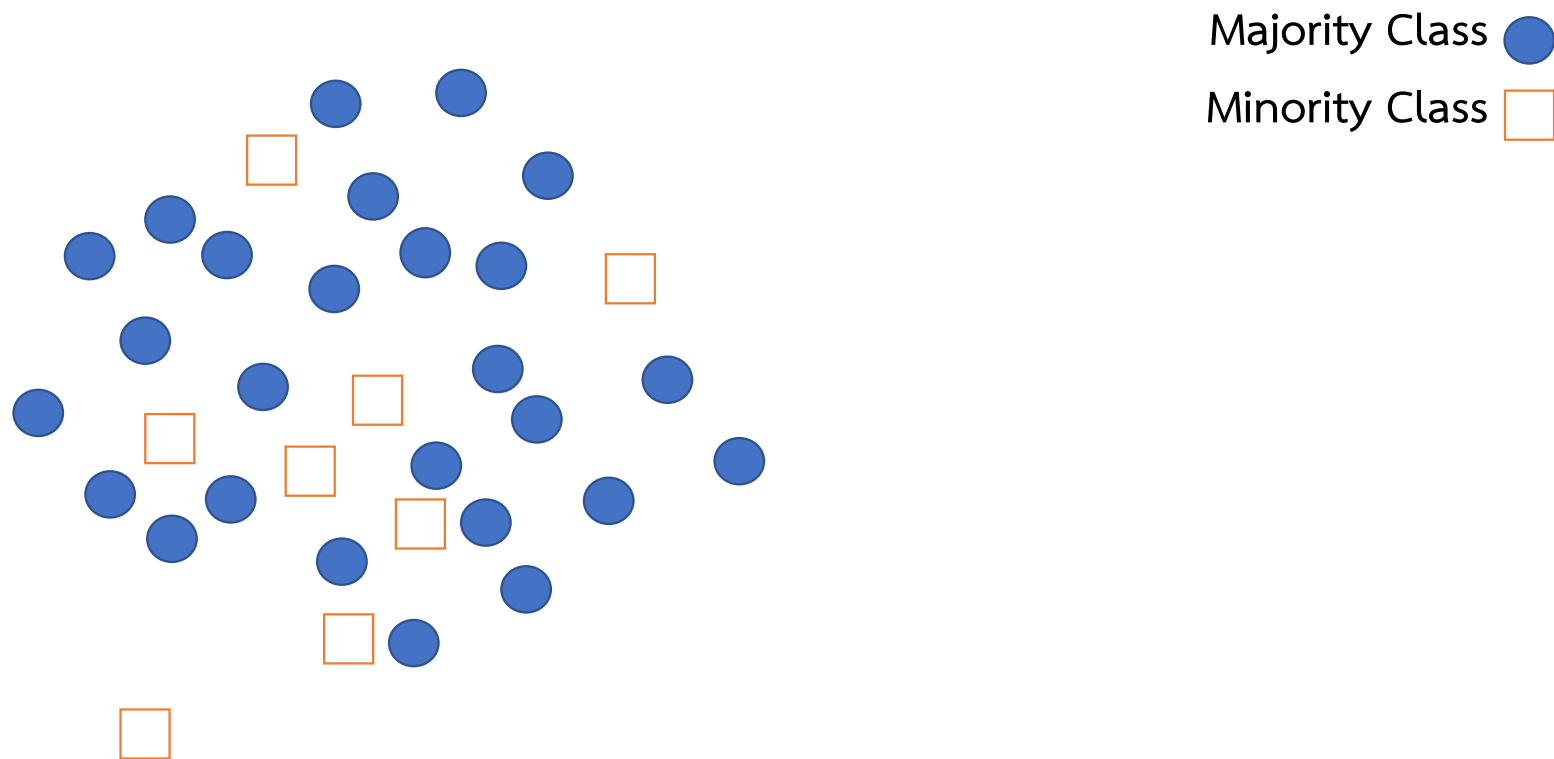
With feature engineering Interaction term

การจัดการกับข้อมูลไม่สมดุล (Imbalanced data)

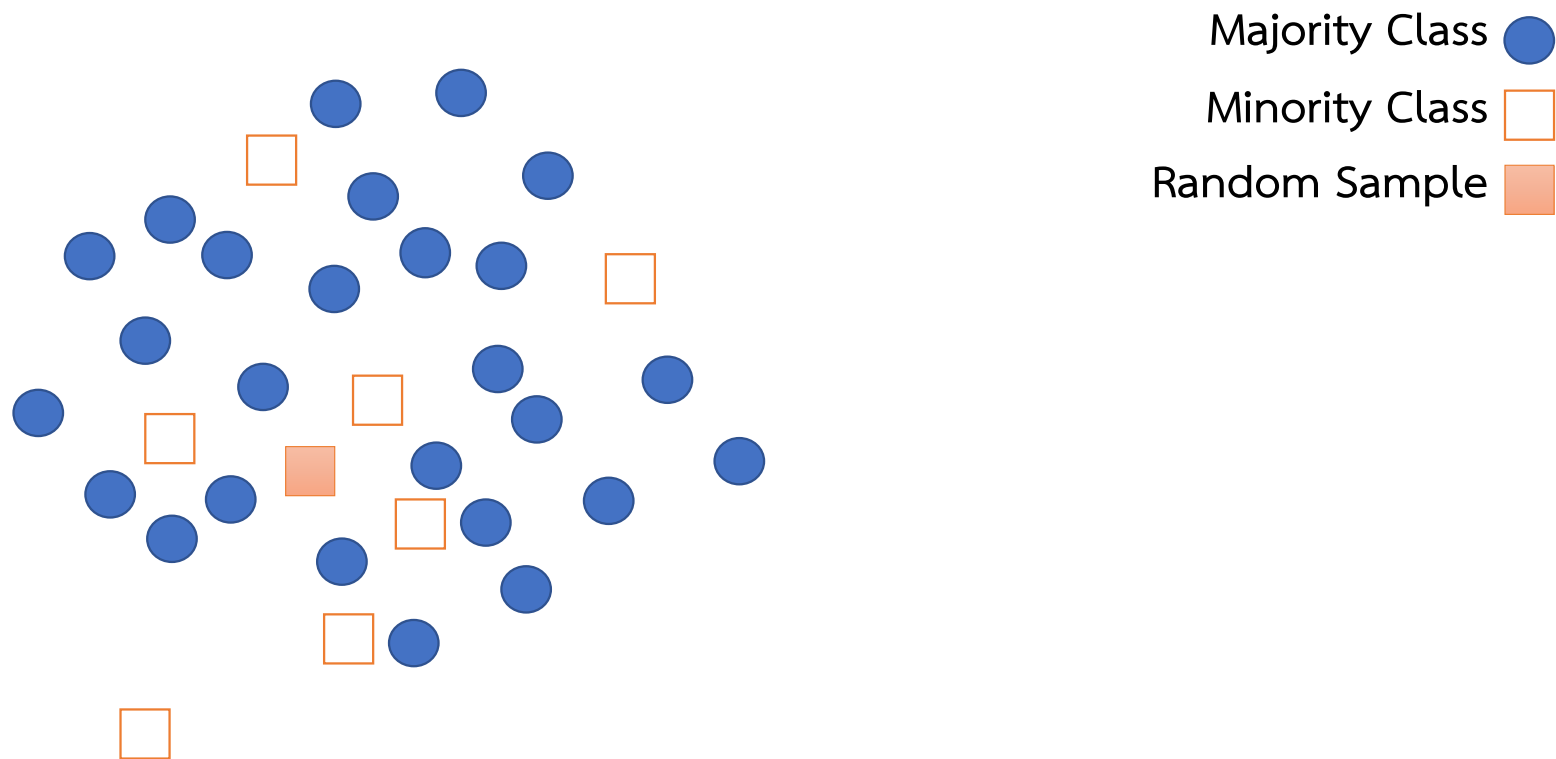


<https://www.kaggle.com/code/rajja/resampling-strategies-for-imbalanced-datasets/notebook>

Synthetic Minority Oversampling Technique (SMOTE)

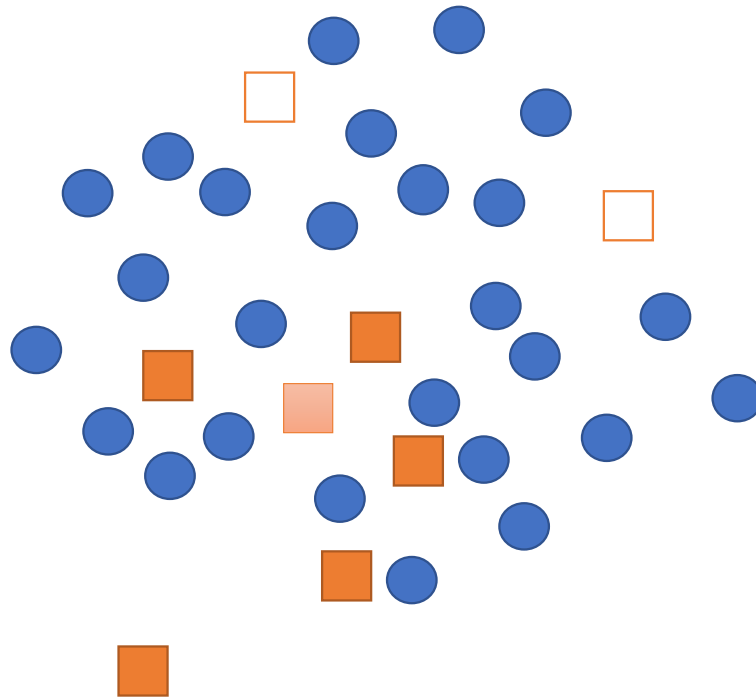


Synthetic Minority Oversampling Technique (SMOTE)



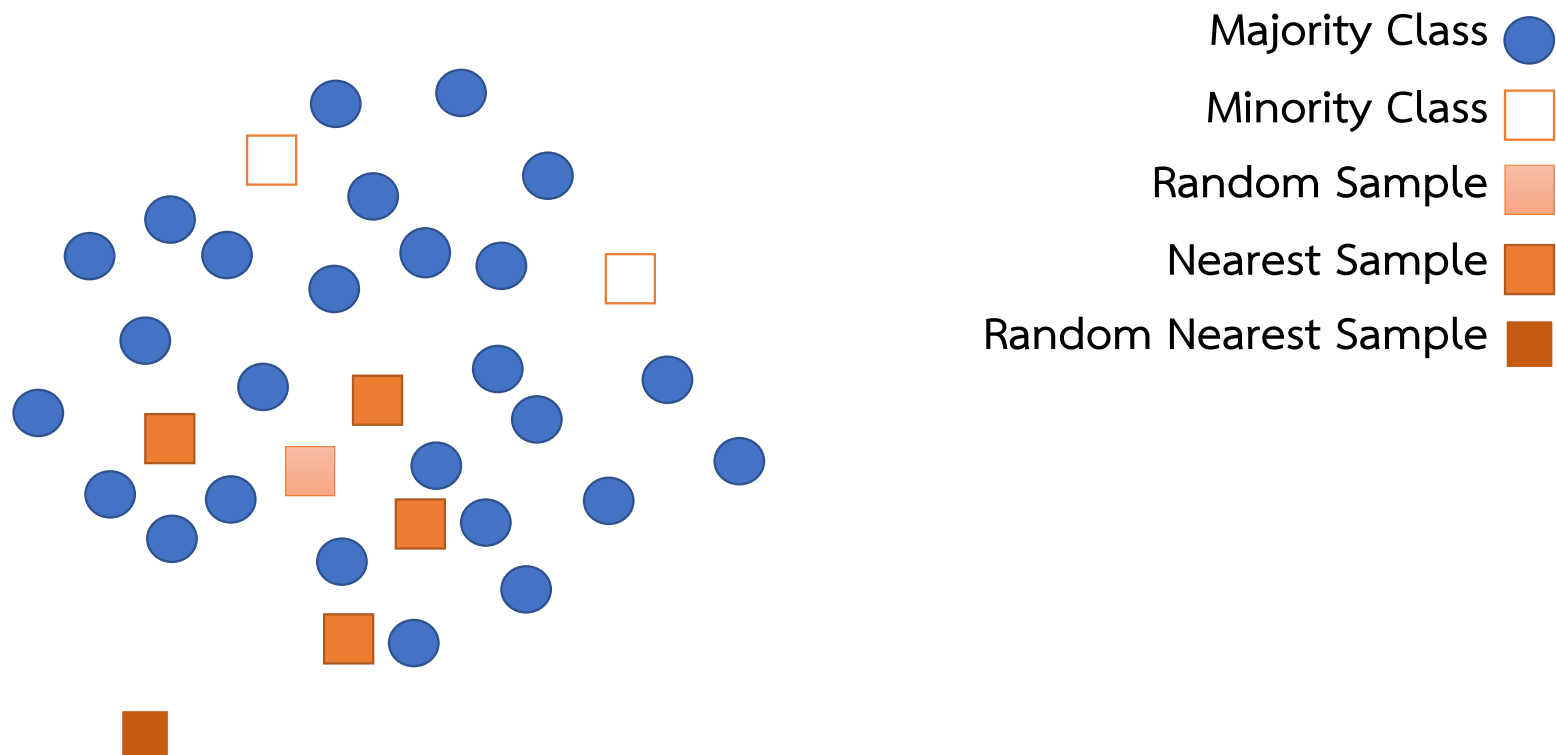
Synthetic Minority Oversampling Technique (SMOTE)

Hyper-parameter (K = 5)

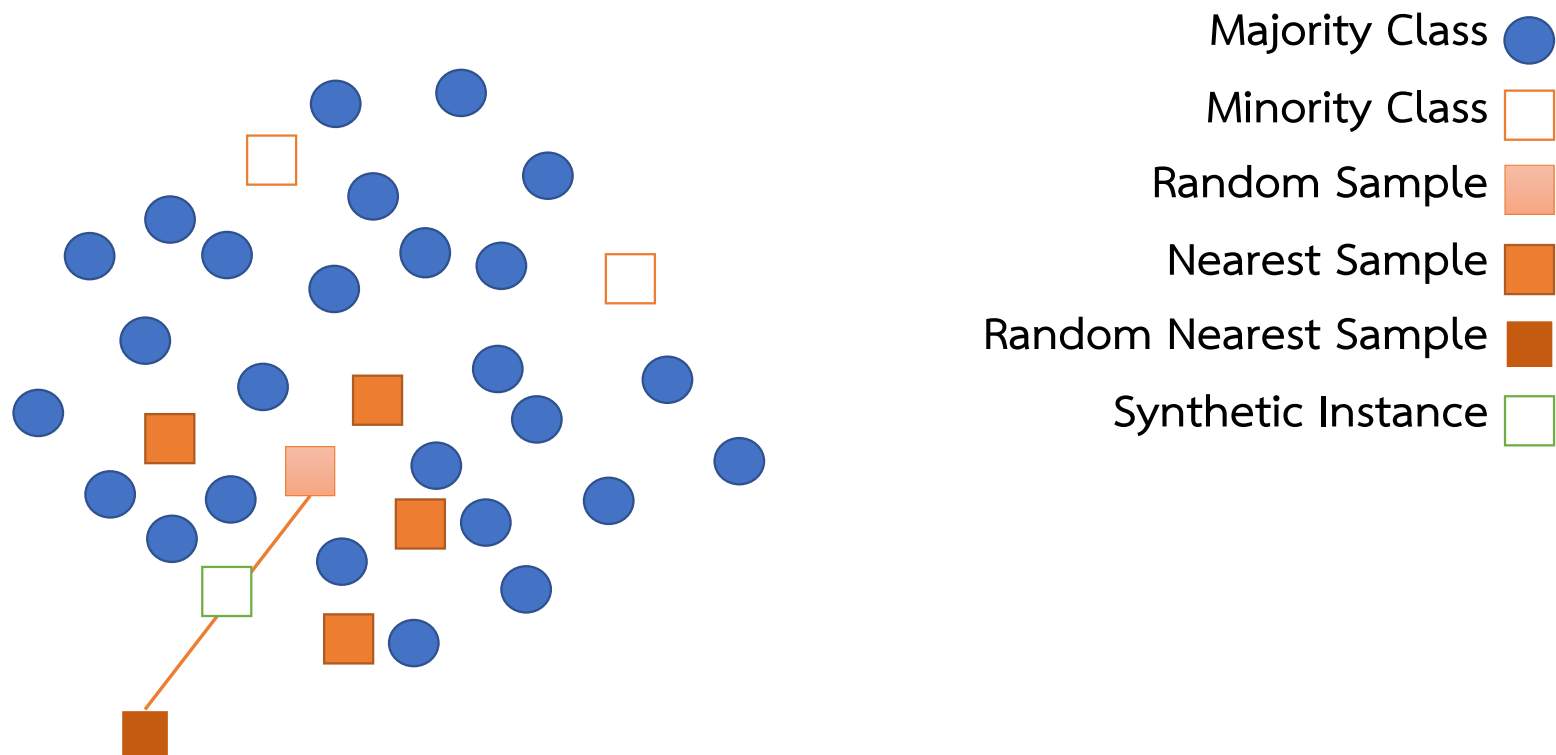


- Majority Class ●
- Minority Class ◻
- Random Sample ◻
- Nearest Samples ■

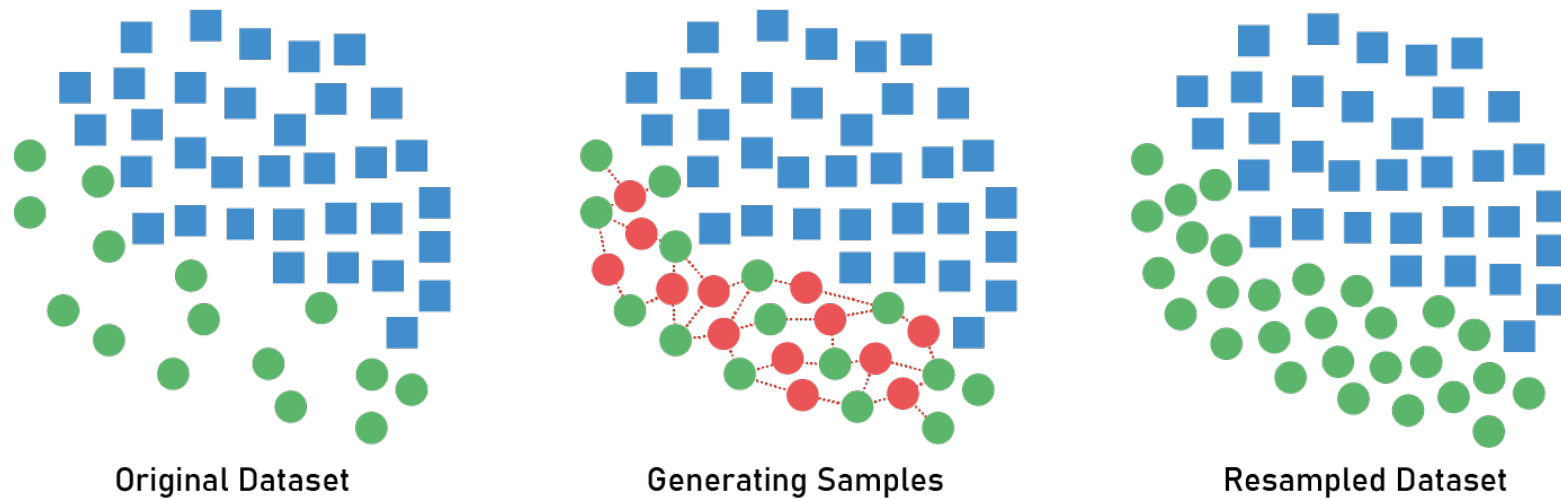
Synthetic Minority Oversampling Technique (SMOTE)



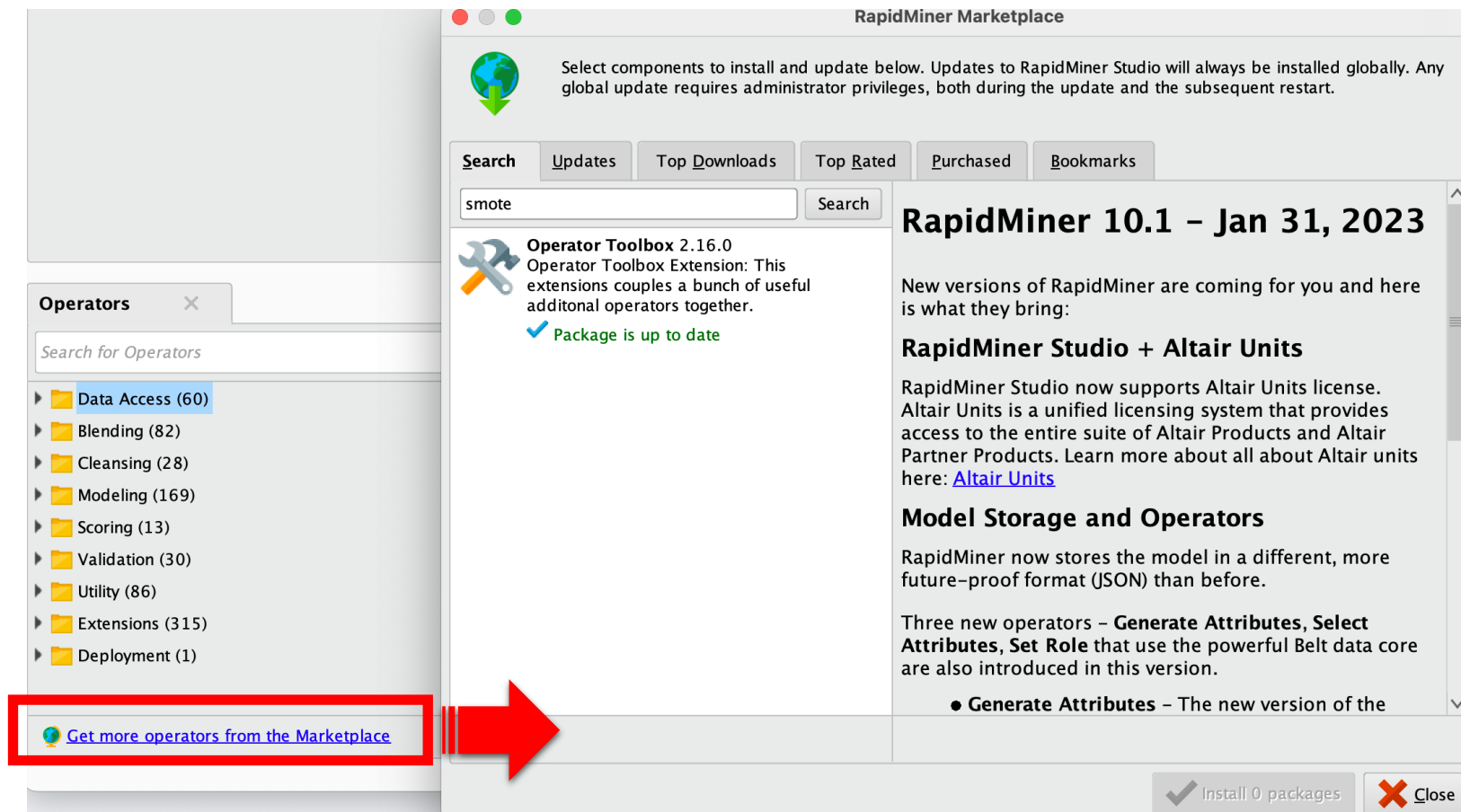
Synthetic Minority Oversampling Technique (SMOTE)



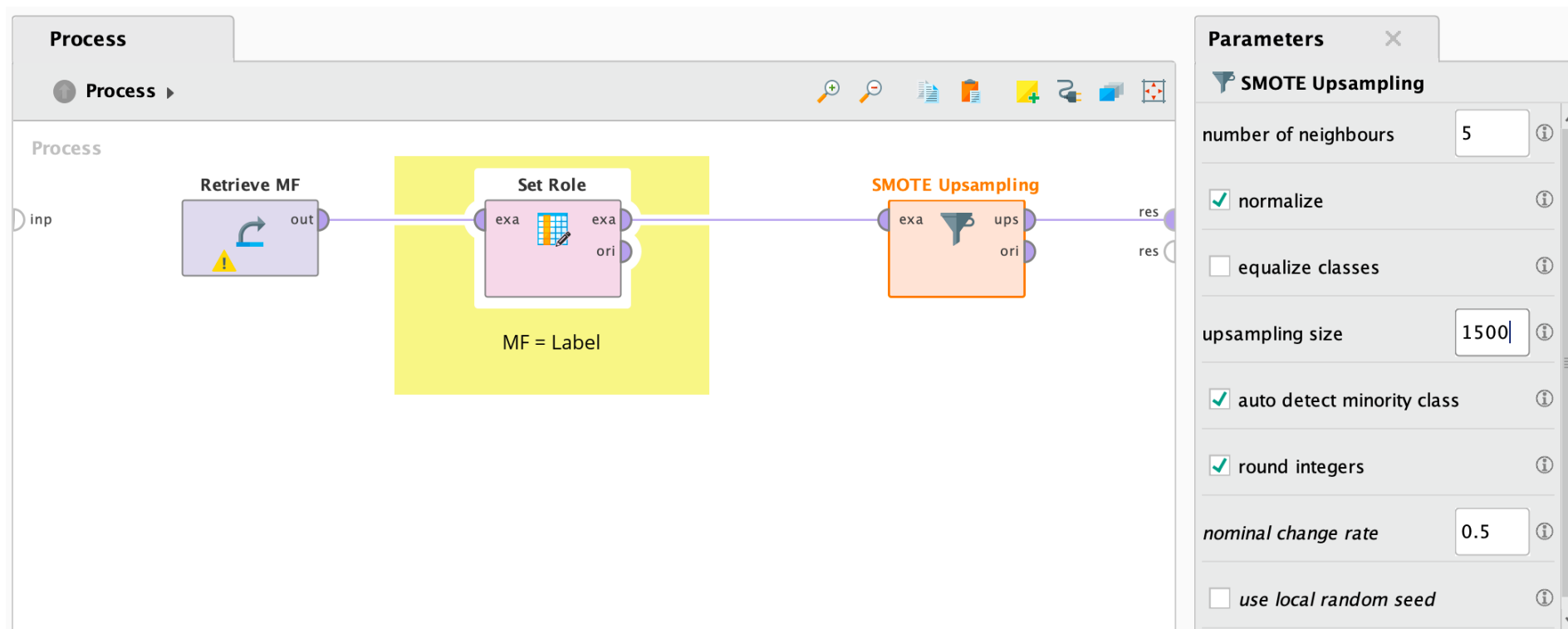
Synthetic Minority Oversampling Technique (SMOTE)



Get more operators from the marketplace

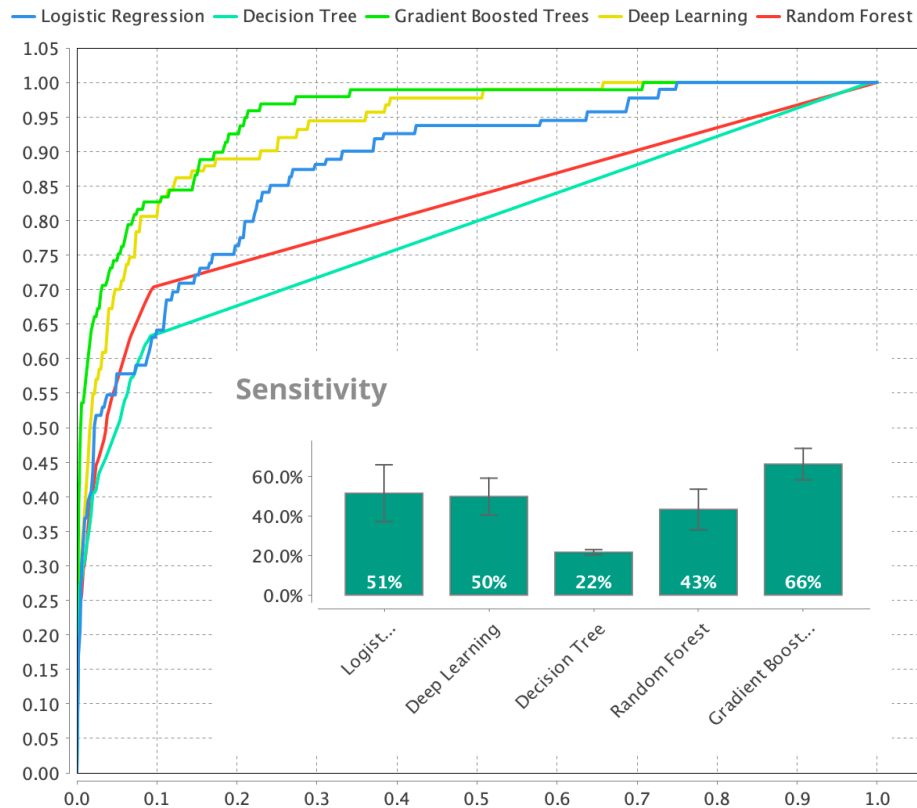


SMOTE Upsampling

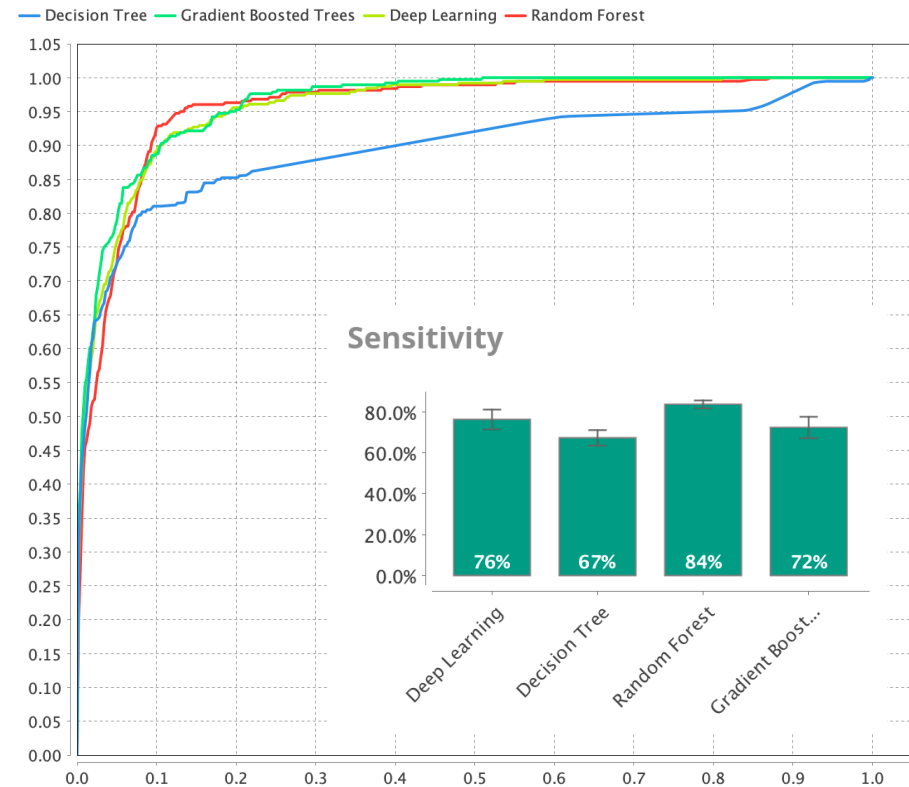


Auto Model – Machine Predictive Maintenance with SMOTE

ROC Comparison



ROC Comparison

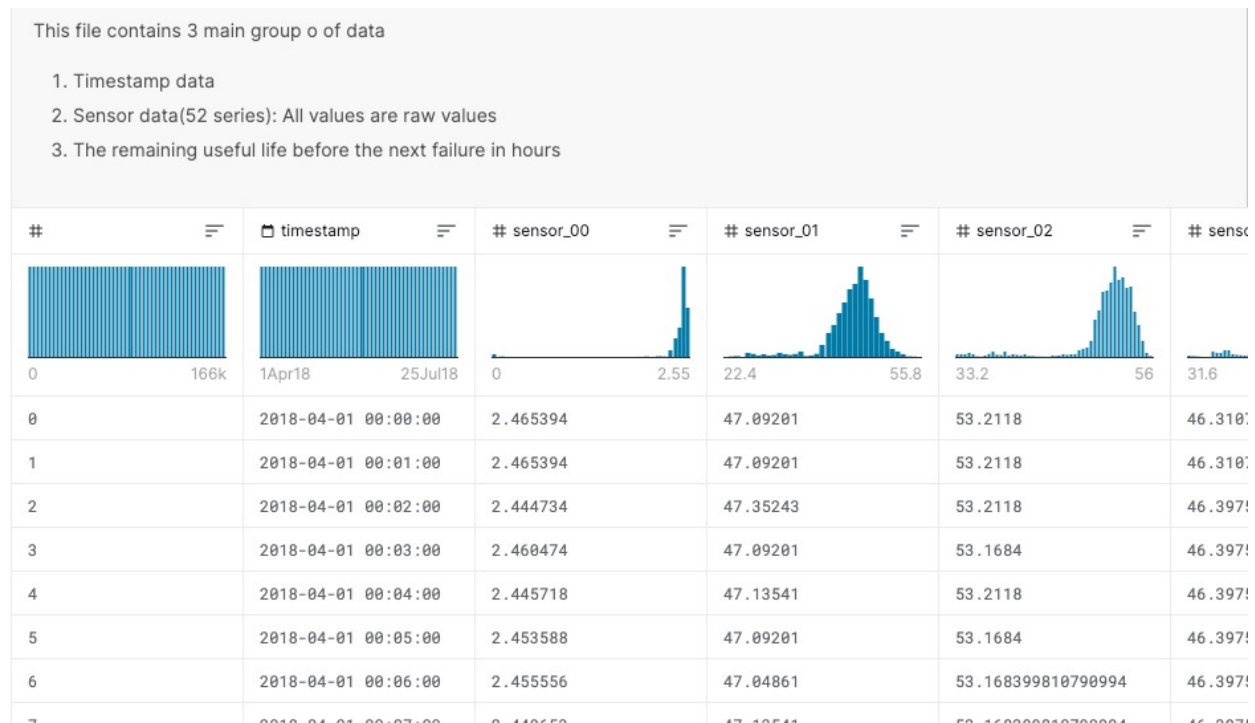


With SMOTE upsampling

Auto Model LAB



- Use case: Water Pump RUL Predictive Maintenance [Regression]
- <https://www.kaggle.com/datasets/anseldsouza/water-pump-rul-predictive-maintenance>
- Dataset: rul_hrs_s.csv

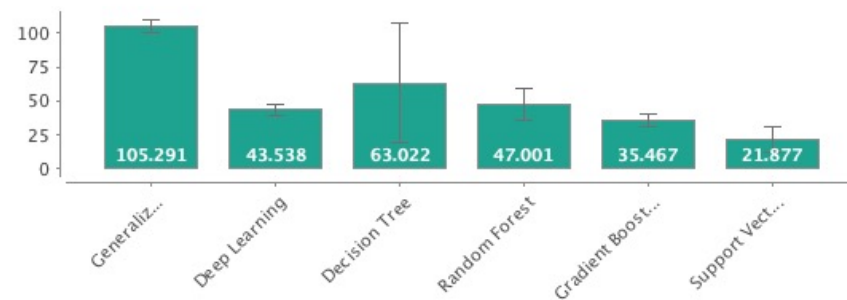


Auto model – Example Result RUL

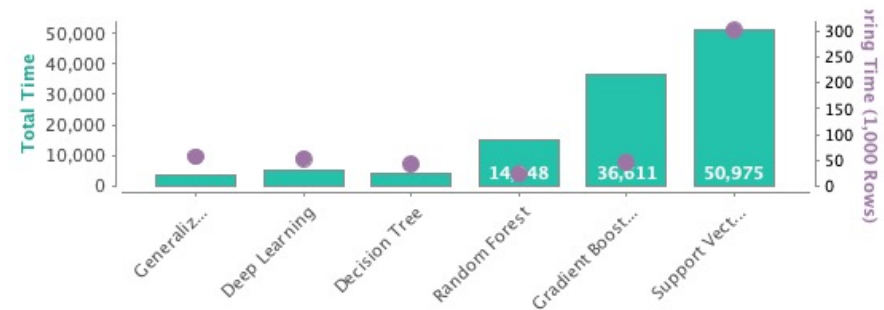
Overview

Number of Models: 8

Root Mean Squared Error



Runtimes (ms)



Root Mean Squared Error ▼

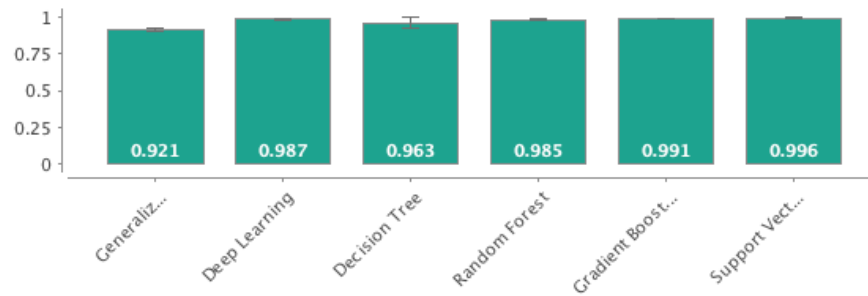
Model		Root Mean Squared...	Standard Deviation	Gains	Total Time	Training Time
Generalized Linear Model	🏃	105.291	± 4.599	?	3 s	101 ms
Deep Learning		43.538	± 4.019	?	5 s	755 ms
Decision Tree		63.022	± 44.215	?	4 s	116 ms
Random Forest	🏃	47.001	± 11.882	?	15 s	61 ms
Gradient Boosted Trees		35.467	± 4.382	?	37 s	399 ms
Support Vector Machine	🏆	21.877	± 9.333	?	51 s	6 s

Auto model – Example Result RUL

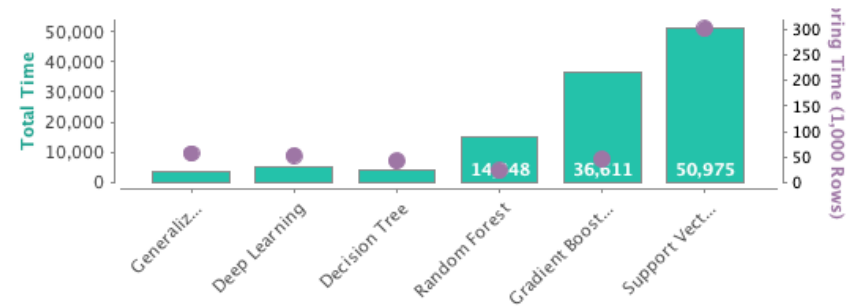
Overview

Number of Models: 8

Correlation



Runtimes (ms)



Correlation ▼

Model		Correlation	Standard Deviation	Gains	Total Time	Training Time
Generalized Linear Model	🚀	0.921	± 0.011	?	3 s	101 ms
Deep Learning		0.987	± 0.003	?	5 s	755 ms
Decision Tree		0.963	± 0.039	?	4 s	116 ms
Random Forest	🚀	0.985	± 0.009	?	15 s	61 ms
Gradient Boosted Trees		0.991	± 0.002	?	37 s	399 ms
Support Vector Machine	🏆	0.996	± 0.003	?	51 s	6 s

Q & A